

Divy Patel

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Education

Arizona State University (ASU), Tempe, AZ

Aug. 2022 – Dec. 2023

Master of Science in Computer Science (MSCS) (GPA: 3.55/4)

Pandit Deendayal Energy University (PDEU), India

Apr. 2018 - May 2022

B.Tech in Information and Communication Technology (GPA: 9.22/10)

Technical Knowledge

Programming Languages: Python, R, C++, C#, SQL, GraphQL

Frameworks: RESTful APIs, Pandas, Seaborn, Matplotlib, Numpy, TensorFlow, PyTorch, OpenCV, Scikit-learn, LangChain

Tool and Technologies: LLM, Version Control, CI/CD, Amazon Web Services (AWS), Docker, Jira, Agile, Scrum

Work Experience

GlobalLogic (Walmart), Dallas, Texas

Software Engineer

Oct. 2024 – Present

- Developed Python/Django APIs enabling external TV control through mobile, HDMI-CEC, and IoT devices.
- Integrated new hardware abstraction layer into existing TV OS components by refactoring legacy interfaces to remove dependency on manufacturer library, ensuring seamless continuity of core features across multiple chipsets.
- Designed and implemented integration test interfaces for system-level validation, using BDD with Behave and Python unit tests.
- Diagnosed and resolved bugs across multiple OS components to support QA and production releases.
- Developing an automated defect tracking system to improve reporting efficiency and **save over 6 hours weekly** for the team lead.

PetSmart, Phoenix, Arizona

Software Developer

May 2023 – Oct 2024

- Developed Python and Django-based microservices, deployed on Docker and Kubernetes on AWS and implemented CI/CD pipelines.
- Improved transaction speed by **15%** and data retrieval speed by **20%** with MS SQL Server optimizations.
- Revamped API with Flask and GraphQL, boosting query responsiveness.
- Led QA testing with PyTest and Selenium, automating the testing process and reducing the common bugs.

Quartic.ai, Remote (San Jose, CA)

Data Engineer

Jul. 2021 - Jun. 2022

- Implemented data pipeline for pre-processing live data feeds with Python, GraphQL, and Redis with CI/CD integration.
- Optimized the data pipeline to enhance predictive analytics and real-time risk monitoring.
- Developed a live dashboard for risk monitoring using Grafana and ReactJS.

HOPS Healthcare, India

Artificial Intelligence Intern

Apr. 2021 - Dec. 2021

- Achieved **85%** accuracy in mobile-based skin disease detection by implementing OpenCV, TensorFlow and CNNs.
- Developed noise reduction algorithm in Python, aiding for detection of heart conditions from smartphone-recorded stethoscope sound.

Projects

LLM for Document Based Question Answering 

Project

- Developed Large Language Model (LLM) using RAG implementation to interpret PDFs, CSV, and JSON files, with **98%** accuracy.
- Used LangChain for efficient storage and retrieval, ensuring accurate responses by implementing Prompt Engineering.
- Enhanced chat automation processes by testing model performance on diverse and complex queries.

Process Scheduling Visualization Application 

PDEU

- Developed a Java app to compute important metrics for process scheduling algorithms.
- Implemented features for input and visualization of process data, including options to add, edit, and execute scheduling algorithms.
- Generated visual graphs to represent calculated metrics, enhancing user understanding of scheduling performance.

Skin Disease Detector App 

Google DSC Hackathon

- Led the development of skin disease detector app during a 24-hour hackathon, using TensorFlow Lite for real-time classification.
- Preprocessed and augmented Harvard skin disease images using OpenCV and did model tuning to achieve **82%** accuracy.
- Developed and integrated the model into an Android app for disease classification via smartphone camera and to find nearest dermatologist.

Student Drowsiness Detection System 

Project

- Extracted 57,488 images from ultraLDD dataset, applied masking using dlib, faceutils along-with OpenCV to focus on eyes and mouth.
- Optimized data handling with a custom TensorFlow data generator for memory-efficient CNN training, achieving **93%** accuracy.
- Implemented real-time prediction from live camera feeds or saved videos, displaying drowsiness detection results with overlay on the screen

Smart Campus Human Detection

Project

- Annotated a dataset used for a smart campus project involving human detection with a 3D LiDAR camera.
- Developed and tested a Neural Network AI model using 2D and 3D CNNs to detect human presence, achieving **98%** accuracy.
- Contributed to the development of a smart campus solution aimed at monitoring and managing crowd density in campus areas.

Recent Achievements

- Developed data-analysis scripts for research on rhesus macaque genetics submitted to *American Journal of Primatology* Jun. 2024
- Publication in "MaterialsToday: Proceedings" on "[Artificial intelligence powered material search engine](#)" Apr. 2022
- Led the team securing 2nd Prize in hackathon organized by Google's Developer Student Committee Mar. 2021
- Led the team securing 2nd Prize in hackathon organized by Nirma University Mar. 2021